Alexandra Miller

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Education

Northern Arizona University | Flagstaff, AZ *Expected Graduation: Fall 2026*
Bachelor of Science - Mechanical Engineering GPA: *3.8*
Distinctions: Dean's List, Lumberjack Scholarship – full tuition awarded
Courses: Material Science, Engineering Analysis, Thermodynamics, Mechanics of Materials, Numerical Analysis, SOLIDWORKS modeling, Machine Design, Fluids Mechanics

Paradise Valley Community College | Phoenix, AZ *Graduation: Spring 2023*
Arizona General Education Certificate (AGEC) *GPA: 4.00*

Experience

Calculus III Mathematics Tutor | NAU Academic Success Center | Flagstaff, AZ *September 2024 - present*
- Supports Calculus III students with their coursework, including homework and quiz/exam corrections
- Works 10 hours a week; gives a 1 to 2 hour lecture every month for exam preparation
- Creates a welcoming and supportive environment for students to learn and increase their problem-solving skills
- Provides resources for students to find additional math help, student services, and peer support
- Students increase scores by 10-20% throughout semester with tutored support

Volunteer Lab Technician | NAU Bioengineering Devices Lab | Flagstaff, AZ *August 2024 – January 2025*
- Collaborated and received two grants from Jean Shuler Research, totaling $1000
- Tested biocompatible polymer materials under direction of Ph.D. candidate to further experimentation on stents
- Assisted in the calibration of the flow model of brain vessels to further testing on catheters used in stroke treatment

Technical Projects

Off Grid Renewable Energy System – Project Manager *January 2026 – December 2026*
- Analyzed and improved an off-grid renewable energy system for Ponderosa High School (PHS)
- Managed team of five through course and client deliverables
- Six STEAM education lessons to high school students for encouraging STEAM careers

ME386 Solar Panel Tracking Project – Project Manager *August 2025 – December 2025*
- Researched and designed solar panel system with dual-axis rotation for tracking photovoltaic waves
- Project prioritizes optimizing solar capture to increase efficiency from panel units

EGR186 Sustainable Composts – Project Leader *August 2023 – December 2023*
- Created four mini composts using reusable materials: a 2-liter soda bottle, binder clips, and cheese cloth
- The composts are affordable, 10 dollars each, and compactible, suitable for college students in dorm/apartment living
- This project won the WL. Gore acceptance for course section

ME286 Bridge Building STEM kit – Project Manager *August 2025 – December 2025*
- Created a STEM kit involving bridge pieces that were different lengths and could be built together with different angles
- Targeted for high schoolers to demonstrate principles of statics

Club Organizations

Vice President | NAU Energy Club | Flagstaff, AZ *January 2026 - present*
- Organization passionate about renewable energies and continuing research for improved energy
- Take notes, prepare supplies for club meetings, conduct outreach to gain awareness and funding for renewable energy
- Partially organized and participated in plant sale at NAU to raise funding for renewable based energy capstones

Skills

Excel |Teams | Word | SOLIDWORKS | MATLAB